Quantification: 701_z26c1-2
==========================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
---------

| measure | value |
|---|---|
| lacuna count | 36 |
| interior lacuna count | 25 |
| frame-edge lacuna count | 11 |

Interior lacunae only: mean, median and sample SD (n = 25)
----------------------------------------------------------

| measure | kind | mean | median | sd | units |
|---|---|---|---|---|---|
| area_px2 | lacuna shape | 2059.24 | 1422 | 2588.69 | px^2 |
| major_axis_length_px | lacuna shape | 74.114 | 63.8935 | 37.7848 | px |
| minor_axis_length_px | lacuna shape | 36.4083 | 30.2759 | 21.4604 | px |
| aspect_ratio | lacuna shape | 2.1894 | 2.0618 | 0.7626 | unitless |
| eccentricity | lacuna shape | 0.8239 | 0.8745 | 0.1557 | unitless |
| solidity | lacuna shape | 0.7404 | 0.7421 | 0.0904 | unitless |
| roots_count | headline | 8.84 | 8 | 3.9967 | count |
| ring_length_r30_px | headline | 339.6 | 320 | 150.013 | px |
| ring_length_r60_px | reported | 678.16 | 639 | 356.85 | px |
| owned_length_px | ownership-dependent | 967.208 | 682.215 | 829.816 | px |
| edge_count | ownership-dependent | 65.64 | 41 | 55.5847 | count |
| mean_edge_length_px | ownership-dependent | 14.8057 | 14.8796 | 2.0745 | px |
| perimeter_px | normalised | 284.386 | 202.35 | 176.832 | px |
| ring_area_r30_px2 | normalised | 7024.8 | 6742 | 3401.72 | px^2 |
| ring_area_r60_px2 | normalised | 13613.2 | 14762 | 7007.82 | px^2 |
| in_frame_fraction_r30 | normalised | 0.9978 | 1 | 0.0109 | unitless |
| in_frame_fraction_r60 | normalised | 0.9926 | 1 | 0.0294 | unitless |
| ring_density_r30 | normalised | 0.0503793 | 0.0468187 | 0.0133018 | px^-1 |
| ring_density_r60 | normalised | 0.0513669 | 0.0464184 | 0.0143932 | px^-1 |
| roots_per_100px_perimeter | normalised | 3.5519 | 3.4111 | 1.1159 | per 100 px |
| ring_attached_length_r30_px | network v2 | 292.96 | 262 | 138.431 | px |
| ring_attached_length_r60_px | network v2 | 504.52 | 453 | 318.538 | px |
| ring_length_w_r30_px | network v2 | 382.954 | 359.208 | 169.144 | px |
| ring_length_w_r60_px | network v2 | 763.276 | 731.997 | 403.053 | px |
| sholl_crossings_r10 | network v2 | 8.68 | 7 | 4.3753 | count |
| sholl_crossings_r20 | network v2 | 8.76 | 8 | 4.2942 | count |
| sholl_crossings_r30 | network v2 | 8.32 | 8 | 4.9977 | count |
| loop_count | loops, report only | 4.08 | 1 | 6.3765 | count |
| loop_edge_share | loops, report only | 0.1481 | 0.1528 | 0.1608 | share |
| ring_width_mean_r30_px | width | 6.0442 | 6.2203 | 1.489 | px |

Per field
---------

| measure | value | units |
|---|---|---|
| analysed_area_px2 | 897823 | px^2 |
| total_skeleton_length_px | 43540 | px |
| canalicular_length_density_per_px | 0.0484951 | px^-1 |
| junction_count | 1260 | count |
| junction_density_per_px2 | 0.00140339 | px^-2 |
| lacuna_count | 36 | count |

| | | |
|---|---|---|
| lacunae_per_px2 | 4.01e-05 | px^-2 |
| skeleton_component_count | 684 | count |
| mean_component_length_px | 63.655 | px |
| median_component_length_px | 14 | px |
| field_length_density_w_per_px | 0.0545299 | px^-1 |
| field_density_without_flagged_per_px | 0.0489925 | px^-1 |
| field_density_in_roi_per_px | | px^-1 |
| unattached_length_px | 286 | px |
| unattached_component_count | 16 | count |
| canalicular_length_density_attached_per_px | 0.0484951 | px^-1 |
| field_length_density_with_unattached_per_px | 0.0488136 | px^-1 |
| loop_count | 161 | count |
| loop_edge_share | 0.2128 | share |
| edge_count | 3186 | count |

Canalicular width, the new measure

| measure | value | units |
|---|---|---|
| width_mean_px | 6.4749 | px |
| width_median_px | 6 | px |
| width_p10_px | 2.8284 | px |
| width_p90_px | 10.198 | px |
| width_pixels_used | 29087 | count |

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

| lacuna_id | area_px2 | major_axis_length_px | minor_axis_length_px | aspect_ratio | eccentricity | solidity | orientation_rad | centroid_row_px | centroid_col_px | on_border |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 6622 | 126.492 | 79.7855 | 1.5854 | 0.776 | 0.7588 | 0.7952 | 43.4012 | 272.285 | True |
| 2 | 24115 | 285.31 | 118.544 | 2.4068 | 0.9096 | 0.7591 | 1.2945 | 92.9897 | 132.363 | True |
| 3 | 8121 | 199.992 | 79.1327 | 2.5273 | 0.9184 | 0.6393 | 0.943 | 189.47 | 54.1918 | True |
| 4 | 6792 | 139.701 | 72.9543 | 1.9149 | 0.8528 | 0.7645 | 1.2651 | 31.6926 | 963.876 | True |
| 5 | 2287 | 125.157 | 32.9459 | 3.7989 | 0.9647 | 0.6065 | 0.5498 | 75.4101 | 463.222 | False |
| 6 | 3999 | 122.318 | 44.0398 | 2.7774 | 0.9329 | 0.877 | 0.1379 | 179.418 | 1003.78 | True |
| 7 | 1531 | 85.4004 | 40.5987 | 2.1035 | 0.8798 | 0.5916 | 1.0809 | 183.008 | 698.054 | False |
| 8 | 2493 | 118.127 | 38.0041 | 3.1083 | 0.9468 | 0.6054 | -1.3995 | 230.773 | 666.071 | False |
| 9 | 5110 | 133.797 | 67.4988 | 1.9822 | 0.8634 | 0.6757 | 1.2061 | 267.96 | 890.703 | False |
| 10 | 1782 | 62.6142 | 47.9495 | 1.3058 | 0.6431 | 0.682 | 1.1297 | 243.333 | 206.608 | False |
| 11 | 459 | 45.5099 | 13.5856 | 3.3499 | 0.9544 | 0.8438 | 0.9355 | 286.754 | 149.582 | False |
| 12 | 1422 | 68.8124 | 28.5123 | 2.4134 | 0.9101 | 0.8335 | 1.1259 | 314.225 | 198.443 | False |
| 13 | 1652 | 67.0082 | 31.8663 | 2.1028 | 0.8797 | 0.8959 | 0.9049 | 359.245 | 256.938 | False |
| 14 | 2608 | 67.6663 | 54.9533 | 1.2313 | 0.5835 | 0.8644 | 0.6706 | 338.284 | 24.3976 | True |
| 15 | 30788 | 334.972 | 147.951 | 2.2641 | 0.8972 | 0.683 | 0.7858 | 446.787 | 150.129 | True |
| 16 | 12693 | 158.381 | 111.247 | 1.4237 | 0.7118 | 0.8419 | -1.4075 | 620.143 | 187.231 | False |
| 17 | 967 | 67.0717 | 21.9089 | 3.0614 | 0.9451 | 0.7662 | -1.3191 | 319.159 | 644.63 | False |
| 18 | 8889 | 153.266 | 79.6369 | 1.9246 | 0.8544 | 0.9019 | 1.3702 | 455.699 | 960.443 | True |
| 19 | 821 | 63.8935 | 20.0569 | 3.1856 | 0.9495 | 0.7609 | 0.1039 | 594.211 | 741.577 | False |
| 20 | 4845 | 153.224 | 58.0129 | 2.6412 | 0.9256 | 0.6393 | -0.5 | 676.391 | 794.742 | False |
| 21 | 2272 | 79.8087 | 42.6154 | 1.8728 | 0.8455 | 0.7881 | -1.0855 | 686.797 | 269.451 | False |
| 22 | 640 | 42.0108 | 20.3754 | 2.0618 | 0.8745 | 0.8625 | 1.0676 | 716.059 | 885.861 | False |
| 23 | 656 | 33.1626 | 30.2759 | 1.0953 | 0.4081 | 0.801 | 1.2644 | 753.977 | 863.463 | False |
| 24 | 883 | 72.1056 | 21.4511 | 3.3614 | 0.9547 | 0.6761 | -1.2791 | 751.882 | 124.115 | False |

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| 25 | 1013 | 51.2657 | 31.9563 | 1.6042 | 0.7819 | 0.7251 | -0.0492 | 792.697 | 1003.49 | True |
| 26 | 1435 | 61.7195 | 33.8344 | 1.8242 | 0.8363 | 0.8112 | -1.4959 | 787.26 | 645.571 | False |
| 27 | 4376 | 111.203 | 63.8724 | 1.741 | 0.8186 | 0.6554 | -1.0966 | 845.546 | 482.61 | False |
| 28 | 633 | 36.1911 | 26.3602 | 1.3729 | 0.6852 | 0.7421 | -1.1058 | 875.441 | 532.795 | False |
| 29 | 474 | 32.816 | 22.1225 | 1.4834 | 0.7386 | 0.7732 | 0.5755 | 897.675 | 908.236 | False |
| 30 | 1500 | 55.4417 | 51.2017 | 1.0828 | 0.3835 | 0.6268 | -1.2338 | 890.901 | 936.497 | False |
| 31 | 405 | 33.2285 | 16.2908 | 2.0397 | 0.8716 | 0.8544 | -1.2783 | 904.956 | 141.983 | False |
| 32 | 718 | 51.4892 | 21.7937 | 2.3626 | 0.906 | 0.7349 | -1.184 | 917.613 | 557.102 | False |
| 33 | 563 | 40.0099 | 23.9496 | 1.6706 | 0.8011 | 0.7136 | 0.7843 | 936.179 | 478.274 | False |
| 34 | 864 | 54.6671 | 23.8765 | 2.2896 | 0.8996 | 0.7279 | -1.1377 | 941.147 | 439.961 | False |
| 35 | 3941 | 116.706 | 45.969 | 2.5388 | 0.9192 | 0.89 | -1.4794 | 1004.56 | 375.487 | True |
| 36 | 2384 | 124.611 | 32.5415 | 3.8293 | 0.9653 | 0.6932 | -1.5289 | 1010.89 | 775.227 | True |

Network per lacuna

| lacuna_id | roots_count | ring_length_r30_px | ring_length_r60_px | owned_length_px | edge_count | mean_edge_length_px | ring_width_mean_r30_px |
|---|---|---|---|---|---|---|---|
| 1 | 15 | 344 | 618 | 349.421 | 24 | 14.5592 | 5.9536 |
| 2 | 26 | 791 | 905 | 814.482 | 66 | 12.3406 | 4.6531 |
| 3 | 10 | 327 | 384 | 217.995 | 17 | 12.8232 | 5.9014 |
| 4 | 14 | 492 | 731 | 432.676 | 39 | 11.0943 | 5.4987 |
| 5 | 19 | 555 | 1192 | 748.027 | 57 | 13.1233 | 6.735 |
| 6 | 11 | 382 | 548 | 859.11 | 51 | 16.8453 | 6.1919 |
| 7 | 11 | 360 | 613 | 403.706 | 29 | 13.9209 | 5.9174 |
| 8 | 12 | 464 | 847 | 1124.78 | 72 | 15.6219 | 5.0033 |
| 9 | 18 | 567 | 1202 | 2074.78 | 142 | 14.6111 | 6.7551 |
| 10 | 9 | 310 | 387 | 263.823 | 17 | 15.519 | 5.9108 |
| 11 | 5 | 117 | 120 | 179.924 | 10 | 17.9924 | 6.0585 |
| 12 | 6 | 153 | 158 | 168.924 | 10 | 16.8924 | 3.8149 |
| 13 | 4 | 202 | 436 | 98.1543 | 5 | 19.6309 | 4.3661 |
| 14 | 1 | 27 | 84 | 7.8284 | 1 | 7.8284 | 9.1513 |
| 15 | 12 | 520 | 983 | 330.563 | 24 | 13.7735 | 4.4296 |
| 16 | 12 | 382 | 711 | 794.014 | 54 | 14.704 | 5.4258 |
| 17 | 9 | 294 | 549 | 412.191 | 32 | 12.881 | 6.4769 |
| 18 | 7 | 469 | 1252 | 1648.74 | 109 | 15.126 | 5.4301 |
| 19 | 6 | 297 | 639 | 944.95 | 62 | 15.2411 | 7.1772 |
| 20 | 10 | 625 | 1123 | 2187.42 | 130 | 16.8263 | 4.1736 |
| 21 | 8 | 333 | 825 | 1210.64 | 78 | 15.521 | 6.7465 |
| 22 | 5 | 164 | 383 | 151.569 | 17 | 8.9158 | 4.1534 |
| 23 | 7 | 190 | 292 | 277.723 | 22 | 12.6238 | 5.4386 |
| 24 | 8 | 486 | 1127 | 2112.94 | 155 | 13.6319 | 7.8175 |
| 25 | 6 | 190 | 339 | 235.48 | 12 | 19.6234 | 5.8946 |
| 26 | 8 | 487 | 1082 | 2019.02 | 140 | 14.4216 | 7.4873 |
| 27 | 16 | 578 | 1341 | 2221.63 | 158 | 14.061 | 5.938 |
| 28 | 8 | 166 | 244 | 252.953 | 17 | 14.8796 | 2.681 |
| 29 | 4 | 154 | 421 | 682.215 | 41 | 16.6394 | 6.4213 |
| 30 | 8 | 345 | 658 | 363.22 | 23 | 15.7922 | 6.3822 |
| 31 | 5 | 277 | 924 | 2659.97 | 173 | 15.3756 | 6.8562 |
| 32 | 8 | 411 | 843 | 1961.59 | 131 | 14.974 | 8.3854 |
| 33 | 7 | 253 | 379 | 411.534 | 30 | 13.7178 | 8.9378 |
| 34 | 8 | 320 | 458 | 454.492 | 36 | 12.6248 | |
| 35 | 15 | 387 | 728 | 1240.8 | 82 | 15.1317 | 4.193 |
| 36 | 9 | 276 | 681 | 96.3553 | 7 | 13.765 | |

Normalised per lacuna

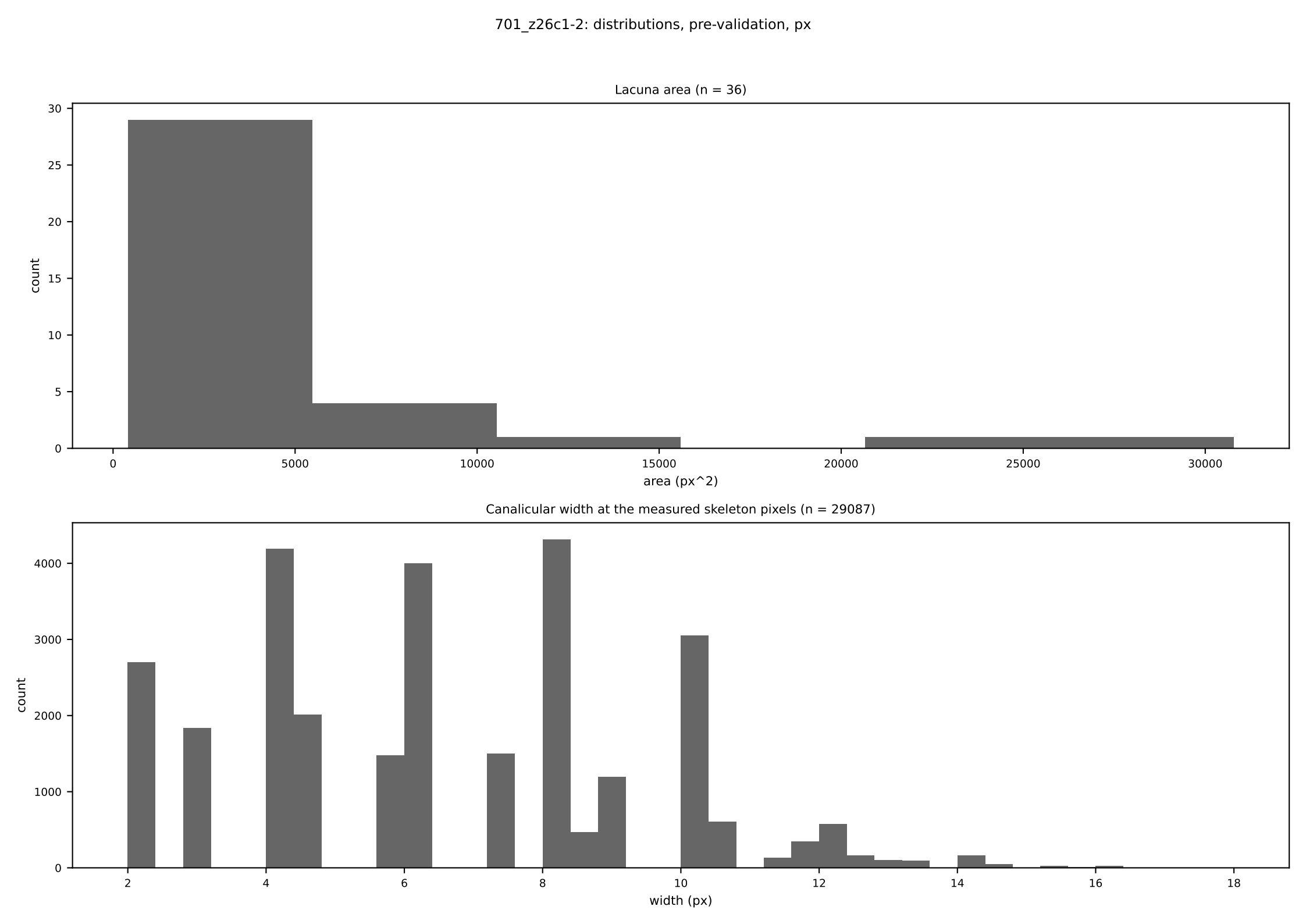
| lacuna_id | perimeter_px | ring_area_r30_px2 | ring_area_r60_px2 | in_frame_fraction_r30 | in_frame_fraction_r60 | ring_density_r30 | ring_density_r60 | roots_per_100px_perimeter |
|---|---|---|---|---|---|---|---|---|
| 1 | 751.134 | 8143 | 12926 | 0.7412 | 0.6831 | 0.0422449 | 0.0478106 | 1.997 |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 2 | 1296.59 | 17823 | 20380 | 0.8324 | 0.6918 | 0.0443809 | 0.0444063 | 2.0053 |
| 3 | 652.802 | 9448 | 10851 | 0.7803 | 0.7374 | 0.0346105 | 0.0353884 | 1.5319 |
| 4 | 561.257 | 7623 | 12937 | 0.5264 | 0.4563 | 0.0645415 | 0.0565046 | 2.4944 |
| 5 | 459.683 | 11984 | 25145 | 0.9457 | 0.8575 | 0.0463117 | 0.047405 | 4.1333 |
| 6 | 376.019 | 7018 | 10546 | 0.6142 | 0.5806 | 0.0544315 | 0.0519628 | 2.9254 |
| 7 | 322.475 | 7086 | 14762 | 1 | 1 | 0.0508044 | 0.0415255 | 3.4111 |
| 8 | 439.713 | 9174 | 16100 | 1 | 1 | 0.0505777 | 0.0526087 | 2.7291 |
| 9 | 541.524 | 14452 | 28872 | 1 | 1 | 0.0392333 | 0.041632 | 3.324 |
| 10 | 345.031 | 6742 | 9970 | 1 | 1 | 0.0459804 | 0.0388165 | 2.6085 |
| 11 | 115.539 | 2499 | 2647 | 1 | 1 | 0.0468187 | 0.0453343 | 4.3275 |
| 12 | 202.35 | 3918 | 4194 | 1 | 1 | 0.0390505 | 0.0376729 | 2.9652 |
| 13 | 196.522 | 5790 | 10562 | 1 | 1 | 0.0348877 | 0.0412801 | 2.0354 |
| 14 | 273.48 | 2279 | 3229 | 0.6705 | 0.6272 | 0.0118473 | 0.0260142 | 0.3657 |
| 15 | 1396.63 | 22077 | 31992 | 0.9271 | 0.8877 | 0.0235539 | 0.0307264 | 0.8592 |
| 16 | 659.572 | 11568 | 19301 | 1 | 1 | 0.0330221 | 0.0368375 | 1.8194 |
| 17 | 188.244 | 7262 | 16365 | 1 | 1 | 0.0404847 | 0.0335472 | 4.781 |
| 18 | 546.701 | 11662 | 25518 | 0.7611 | 0.7175 | 0.0402161 | 0.0490634 | 1.2804 |
| 19 | 163.723 | 6608 | 14115 | 1 | 1 | 0.0449455 | 0.045271 | 3.6647 |
| 20 | 660.199 | 14216 | 24193 | 1 | 1 | 0.0439645 | 0.0464184 | 1.5147 |
| 21 | 273.019 | 7642 | 15822 | 1 | 1 | 0.043575 | 0.0521426 | 2.9302 |
| 22 | 128.202 | 4822 | 9334 | 1 | 1 | 0.0340108 | 0.0410328 | 3.9001 |
| 23 | 133.338 | 4666 | 9184 | 1 | 1 | 0.0407201 | 0.0317944 | 5.2498 |
| 24 | 277.948 | 7921 | 17655 | 1 | 1 | 0.0613559 | 0.0638346 | 2.8782 |
| 25 | 208.386 | 5088 | 9797 | 0.7064 | 0.6319 | 0.0373428 | 0.0346024 | 2.8793 |
| 26 | 215.7 | 7866 | 18883 | 1 | 1 | 0.061912 | 0.0573002 | 3.7089 |
| 27 | 605.222 | 10588 | 19891 | 1 | 1 | 0.0545901 | 0.0674174 | 2.6437 |
| 28 | 146.202 | 2975 | 3820 | 1 | 1 | 0.0557983 | 0.0638743 | 5.4719 |
| 29 | 133.717 | 2752 | 7041 | 1 | 1 | 0.0559593 | 0.0597926 | 2.9914 |
| 30 | 307.298 | 7315 | 14828 | 1 | 0.9574 | 0.0471634 | 0.0443755 | 2.6033 |
| 31 | 96.7107 | 5294 | 16215 | 1 | 1 | 0.0523234 | 0.0569843 | 5.1701 |
| 32 | 167.823 | 5199 | 11007 | 1 | 1 | 0.0790537 | 0.0765876 | 4.7669 |
| 33 | 143.616 | 3510 | 5238 | 1 | 1 | 0.0720798 | 0.0723559 | 4.8741 |
| 34 | 186.279 | 3771 | 5185 | 1 | 1 | 0.0848581 | 0.0883317 | 4.2946 |
| 35 | 332.007 | 6043 | 11546 | 0.5904 | 0.5692 | 0.064041 | 0.0630521 | 4.518 |
| 36 | 413.233 | 6749 | 15845 | 0.5979 | 0.5709 | 0.0408949 | 0.0429789 | 2.178 |

Network v2 per lacuna

| lacuna_id | ring_attached_length_r30_px | ring_attached_length_r60_px | ring_length_w_r30_px | ring_length_w_r60_px | sholl_crossings_r10 | sholl_crossings_r20 | sholl_crossings_r30 |
|---|---|---|---|---|---|---|---|
| 1 | 304 | 368 | 379.563 | 684.215 | 8 | 5 | 8 |
| 2 | 697 | 772 | 886.838 | 1019.55 | 18 | 25 | 5 |
| 3 | 240 | 240 | 368.735 | 432.777 | 11 | 9 | 1 |
| 4 | 434 | 457 | 561.76 | 831.21 | 14 | 9 | 12 |
| 5 | 494 | 692 | 631.843 | 1347.07 | 20 | 10 | 12 |
| 6 | 316 | 488 | 430.09 | 617.357 | 10 | 15 | 7 |
| 7 | 283 | 328 | 409.534 | 689.428 | 9 | 8 | 9 |
| 8 | 407 | 569 | 519.032 | 946.11 | 13 | 12 | 8 |
| 9 | 533 | 915 | 647.127 | 1357.24 | 16 | 18 | 17 |
| 10 | 210 | 235 | 338.865 | 421.563 | 7 | 7 | 3 |
| 11 | 102 | 102 | 132.397 | 135.225 | 2 | 2 | 0 |
| 12 | 117 | 117 | 172.61 | 179.267 | 5 | 3 | 2 |
| 13 | 135 | 159 | 231.108 | 496.002 | 6 | 8 | 5 |
| 14 | 8 | 8 | 30.799 | 97.9117 | 1 | 0 | 3 |
| 15 | 287 | 295 | 583.085 | 1099.74 | 10 | 14 | 13 |
| 16 | 315 | 453 | 429.505 | 793.997 | 10 | 8 | 9 |
| 17 | 218 | 321 | 326.936 | 612.115 | 6 | 7 | 7 |
| 18 | 379 | 801 | 519.475 | 1402.79 | 14 | 16 | 13 |
| 19 | 241 | 492 | 347.363 | 731.997 | 6 | 8 | 12 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 20 | 466 | 861 | 698.985 | 1280.24 | 15 | 17 | 11 |
| 21 | 320 | 575 | 374.492 | 920.997 | 8 | 9 | 11 |
| 22 | 126 | 126 | 181.297 | 429.191 | 4 | 7 | 6 |
| 23 | 190 | 240 | 211.095 | 317.622 | 5 | 7 | 2 |
| 24 | 461 | 1057 | 552.404 | 1284.22 | 9 | 14 | 14 |
| 25 | 146 | 192 | 210.167 | 381.333 | 5 | 3 | 6 |
| 26 | 448 | 935 | 543.576 | 1206.61 | 11 | 15 | 19 |
| 27 | 524 | 1196 | 647.245 | 1491.56 | 15 | 13 | 14 |
| 28 | 142 | 219 | 184.125 | 268.267 | 7 | 2 | 2 |
| 29 | 154 | 333 | 176.539 | 479.818 | 3 | 5 | 4 |
| 30 | 262 | 368 | 394.676 | 736.855 | 9 | 9 | 13 |
| 31 | 244 | 802 | 310.551 | 1049.19 | 7 | 8 | 10 |
| 32 | 405 | 776 | 461.534 | 959.767 | 11 | 11 | 8 |
| 33 | 208 | 287 | 291.794 | 434.191 | 6 | 8 | 4 |
| 34 | 319 | 455 | 359.208 | 513.362 | 7 | 3 | 6 |
| 35 | 360 | 669 | 422.664 | 813.227 | 10 | 10 | 9 |
| 36 | 107 | 107 | 303.007 | 755.269 | 5 | 5 | 13 |


701_z26c1-2: distributions, pre-validation, px
Lacuna area (n = 36)
count
area (px^2)
0
5
10
15
20
25
30
0
5000
10000
15000
20000
25000
30000
Canalicular width at the measured skeleton pixels (n = 29087)
count
width (px)
0
1000
2000
3000
4000
2
4
6
8
10
12
14
16
18